Intelligent Thermal Management

A Predictive Edge-AI Framework for Industrial Maintenance

B.Tech ICT Portfolio Project | 2024

EXECUTIVE SUMMARY

Edge Node

High-fidelity data acquisition using Arduino and DHT22 sensors for real-time monitoring.

Live Stream

Seamless data serialization over USB using PySerial to bridge hardware and AI layers.

AI Predictor

Forecasting thermal trajectories using Linear Regression to detect anomalies proactively.

THE CRITICAL RISK

25%

Hardware Failure Rate

The Complication

Traditional monitoring systems are **reactive**. They alert users only after a server room breaches critical limits, causing irreversible component degradation and costly downtime.

This project solves the "Reactive Lag" by introducing predictive forecasting at the edge.

HYBRID ARCHITECTURE

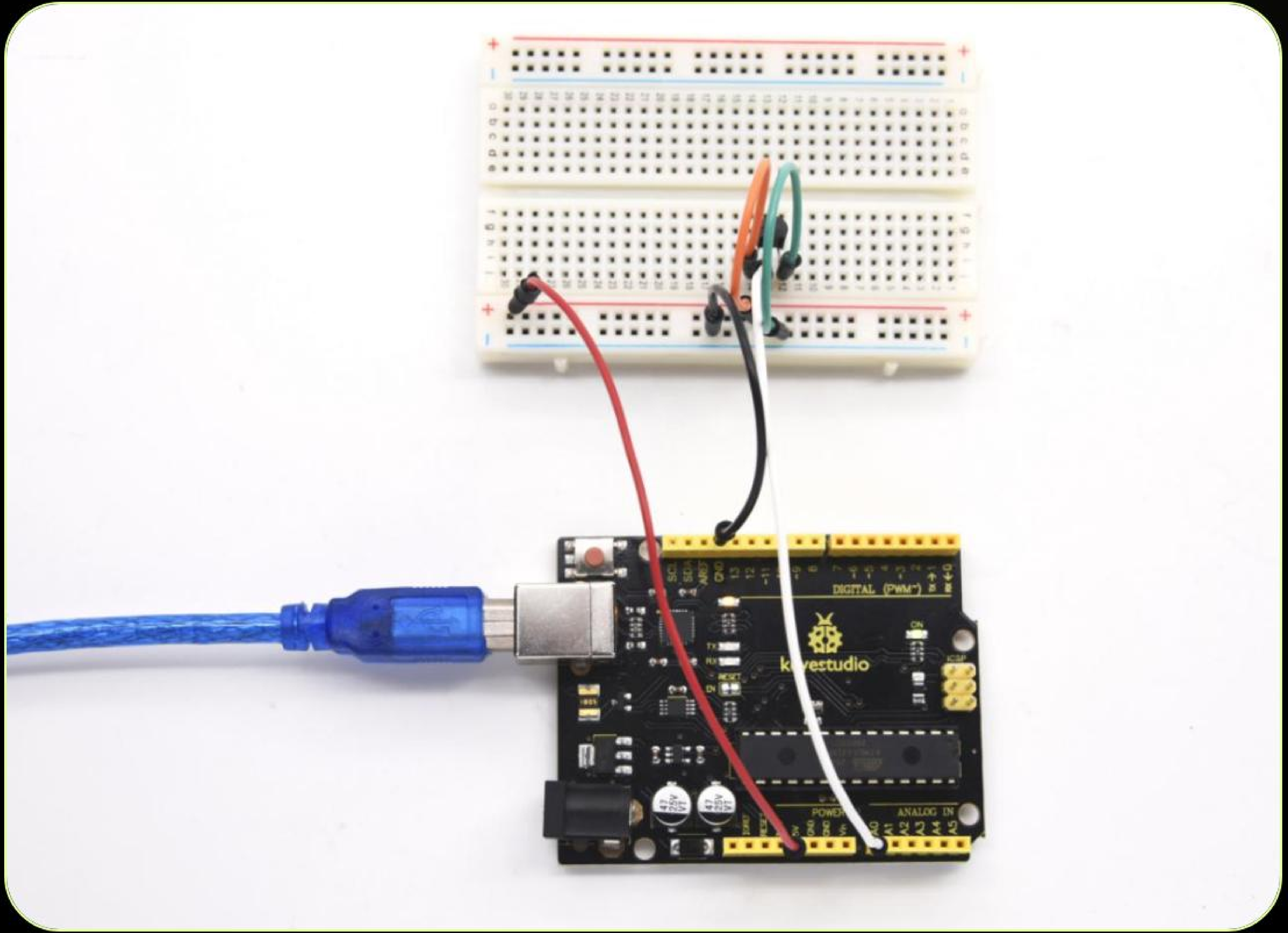
Split-Computing Strategy

Our resolution involves a dual-layer framework. The **Arduino** handles mission-critical safety loops, while the **Python Central Hub** processes complex mathematical models.

This ensures that even if the AI script disconnects, the hardware remains safe via hardwired threshold triggers.

INDUSTRIAL EDGE NODE

- **Arduino Uno:** ATmega328P based edge controller for low-latency sampling.
- **DHT22 Sensor:** High-accuracy thermistor with a 0-100% humidity range.
- **Safety Interface:** Integrated LED array and 1kHz buzzer for local alerts.



LOCAL SAFETY LOGIC

1. Sampling

The DHT22 is queried every 2000ms to ensure reading stability and accuracy.

2. Serialization

Data is packed into CSV format `(T, H)` for optimized serial transmission.

3. Trip Controls

Hardware trips (LED/Buzzer) activate instantly if Temp > 30.0°C independently of AI.

The Predictive Layer

Transitioning from static thresholds to dynamic forecasting.

LINEAR REGRESSION AI

Supervised Training

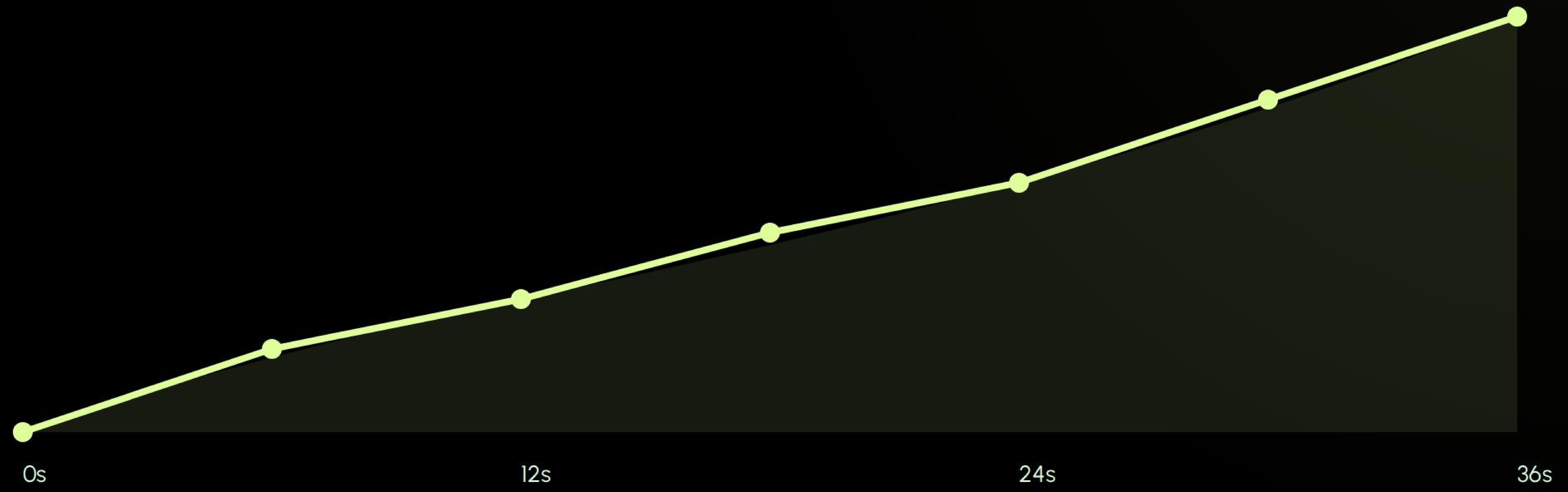
The AI trains on a moving history window. It solves the Least Squares Equation to optimize weights based on real-time room dynamics.

$$\hat{y} = \beta_0 + \sum (\beta_i x_i)$$

Temporal Inference

By mapping the last 3 data steps, the model predicts the temperature 6 seconds in advance, identifying slopes and thermal acceleration patterns.

FORECASTING ACCURACY



Blue line indicates AI predictive trajectory; Alerts fire 6-10 seconds before threshold breach.

RESPONSE TIME EFFICIENCY

SYSTEM SCALABILITY

IoT Mesh

Replacing Serial with MQTT/Wi-Fi using ESP32 nodes for distributed facility monitoring.

Cloud Sync

Integrating AWS/Azure IoT hubs for global data aggregation and long-term trend analysis.

Advanced ML

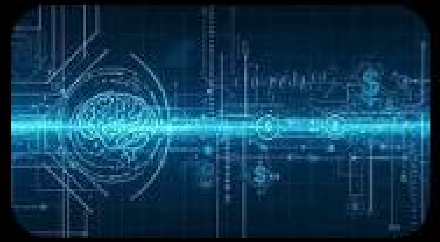
Transitioning from Linear Regression to LSTM Neural Networks for complex non-linear trends.

Questions?

Thank you for your time.

Predictive Industrial Thermal Hub | GitHub Repository

IMAGE SOURCES



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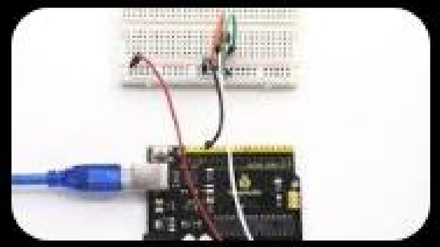
Source: pngtree.com



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Source: docs.keyestudio.com